but if you can make matrix stuff easy.

Abstract vector spaces

- what are vectors?

- once you get comfortable changing basis, you're working independ of coordinate system, it's independant of what basis vectors u chose

both indep of ur coordinate system

- det → (how much L() scales areas)
- eigenvectors → ($\vec{v}$s that stay on own span after L())

. if vectors not just lists of #s, but something more spatial, then whats space?

- functions are another vector
- u can add : $f(x) + g(x) = (f+g)(x)$
- like adding vectors coordinate by coordinate ....
- but functions have infinite coordinates

- u can scale a curve by a # too

- $\therefore$ u can apply these to functions to
  - → linear transformations
  - → null space
  - → dot products
  - → eigen everything

- ex: derivative takes 1 function into another
- what does it mean for transformation of functions to be linear

- we need an abstract def of linearity, to generalize to functions & arrows

additivity: $L(\vec{v} + \vec{w}) = L(\vec{v}) + L(\vec{w})$
scaling: $L(c\vec{v}) = cL(\vec{v})$

"linear trans. preserve addition & scal mul"

visually its grid lines evenly spaced

- lin tran completely described by where it takes the basis vectors

- Derivative is linear.

- Describe derivative as a matrix.
- Current space: All polynomials
- we need coordinates for this space
  → choose a basis.
- polynomials already written as linear combinations...
- So use base powers of x as basis functions

$b_0(x) = 1$
$b_1(x) = x$
$b_2(x) = x^2$
$\vdots$
∞

So when we treat functions as vectors, they'll have infinite coordinates

- in this coordinate system, derivative described by. .

$$\frac{d}{dx} = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 2 & 0 & \cdots \\ 0 & 0 & 0 & 3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

try it: $\frac{d}{dx}\left(1x^3 + 5x^2 + 4x + 5\right)$

$$\frac{d}{dx}\begin{bmatrix} 5 \\ 4 \\ 5 \\ 1 \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 \\ 2 \cdot 5 \\ 3 \cdot 1 \\ \\ \end{bmatrix}$$

constant is 4. bc $\frac{d}{dx}(4x) = 4$
x coeff is 10. bc $\frac{d}{dx}(5x^2) = 10$
x^2 coeff is 1. bc $\frac{d}{dx}(x^3) = 1$

- derivative is linear.
- you can construct $\frac{d}{dx}$ matrix, by taking derivative of each basis function & putting coords of result in each column.

- Linear Transformations

matrix-vector multiplication $\Leftrightarrow$ $\frac{df}{dx}$

- lin tran → linear operators
  dot product → inner product
  eigenvectors → eigen functions

- what is a vector?

anything w/ scaling & adding, you can apply this stuff

- everything vectorish: arrows, lists of #s, functions are vector spaces

- 8 axioms that any vector space must satisfy, to use any Lin alg tool on it

- checklist for vector spaces to satisfy

- mathematician doesnt answer what are vectors? He just checks for notion of adding & scaling that follows axioms

ex: what is 3?

→ concretely: some triplet

→ in math: abstraction for all possible triplets of things, lets u reason w/ all triplets in single idea

- vectors have many embodiments but we abstract them all into a vector space

Done!